MySQL PITR Using mysqldump

Date : 05-09-2026

Steps :

1. Create database
2. Create table
3. Insert a data into that table
4. Take full backup
5. Insert few more records
6. Accedently user delete all data they forgot to apply filter where clause .
7. Restore the full backup
8. Check till where data should be available
9. Apply binary (binlog) file using start-stop time or start-stop position
10. Check the data availability

```
mysql> create database recovery;
Query OK, 1 row affected (0.05 sec)
```

```
mysql> create table recovery_bank(id int primary key auto_increment , name varchar(100), insert_date datetime default current_timestamp);
Query OK, 0 rows affected (0.38 sec)
```

```
mysql> insert into recovery_bank (name) values('HDFC');
Query OK, 1 row affected (0.00 sec)

mysql> insert into recovery_bank (name) values('ICICI');
Query OK, 1 row affected (0.00 sec)

mysql> insert into recovery_bank (name) values('SBI');
Query OK, 1 row affected (0.00 sec)

mysql> insert into recovery_bank (name) values('BANK OF BARODA');
Query OK, 1 row affected (0.00 sec)

mysql> insert into recovery_bank (name) values('UNION BANK OF INDIA');
Query OK, 1 row affected (0.00 sec)
```

```
root@db-03:/recovery_data# mysqldump -u root -p -h localhost recovery >recovery_full_backup.sql;
Enter password:
```

```
mysql> use recovery;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> insert into recovery_bank (name) values('Kenra Bank');
Query OK, 1 row affected (0.00 sec)

mysql> insert into recovery_bank (name) values('Axis Bank');
Query OK, 1 row affected (0.00 sec)

mysql> insert into recovery_bank (name) values('Bank of India');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> select * From recovery_bank;
+----+--------------------+---------------------+
| id | name               | insert_date         |
+----+--------------------+---------------------+
|  1 | HDFC               | 2026-09-05 16:07:47 |
|  2 | ICICI              | 2026-09-05 16:07:54 |
|  3 | SBI                | 2026-09-05 16:08:01 |
|  4 | BANK OF BARODA     | 2026-09-05 16:08:13 |
|  5 | UNION BANK OF INDIA| 2026-09-05 16:08:32 |
|  6 | Kenra Bank         | 2026-09-05 16:11:06 |
|  7 | Axis Bank          | 2026-09-05 16:11:16 |
|  8 | Bank of India      | 2026-09-05 16:11:26 |
+----+--------------------+---------------------+
8 rows in set (0.00 sec)
```

```
mysql> delete from recovery_bank;
Query OK, 8 rows affected (0.00 sec)
```

Steps to recovery the data .

1. Restore the full database backup .

```
root@db-03:/recovery_data# mysql -u root -p -h localhost recovery < recovery_full_backup.sql;
Enter password:
root@db-03:/recovery_data#
```

2. Run to see the data till where it can be reover

```
Database changed
mysql> select * from recovery_bank;
+----+--------------------+---------------------+
| id | name               | insert_date         |
+----+--------------------+---------------------+
|  1 | HDFC               | 2026-09-05 16:07:47 |
|  2 | ICICI              | 2026-09-05 16:07:54 |
|  3 | SBI                | 2026-09-05 16:08:01 |
|  4 | BANK OF BARODA     | 2026-09-05 16:08:13 |
|  5 | UNION BANK OF INDIA| 2026-09-05 16:08:32 |
+----+--------------------+---------------------+
5 rows in set (0.00 sec)
```

Note : in full backup restore we can see the table 5 row data has been recover but still 3 are missing so go to the next step .

3. Now apply the binlog file (check in binlog file)

I have check the binary log and below are the binary log to see that insert and delete command logged .

```
root@db-03:/var/lib/mysql# mysqlbinlog -v mysql-bin.000037
```

After this I have read **specific position** whre it's start from id 6 to till 8

```
root@db-03:/var/lib/mysql#
root@db-03:/var/lib/mysql# mysqlbinlog -v  --start-position=2403 --stop-position=3480  /var/lib/mysql/mysql-bin.000037
```

Now this binlog we want to apply into the mysql database .
First take into seprate file

root@db-03:/var/lib/mysql# mysqlbinlog --base64-output=DECODE-ROWS --start-position=2403 --stop-position=3401 /var/lib/mysql/mysql-bin.000037 > /tmp/recovery.sql

```
root@db-03:/tmp#
root@db-03:/tmp# mysqlbinlog   --base64-output=DECODE-ROWS   --start-position=2403   --stop-position=3401   /var/lib/mysql/mysql-bin.000037 > /tmp/recovery.sql
```

Because in this GTID enable so we need to remove gtid otherwise getting error so remove gtid and apply
root@db-03:/tmp# mysqlbinlog --skip-gtids --start-position=2403 --stop-position=3401 /var/lib/mysql/mysql-bin.000037 > /tmp/table_recovery.sql

```
root@db-03:/tmp#
root@db-03:/tmp# mysqlbinlog   --skip-gtids   --start-position=2403   --stop-position=3401   /var/lib/mysql/mysql-bin.000037 > /tmp/table_recovery.sql
```

Or Restore using specific time after full backup restore :

```
mysqlbinlog \
 --skip-gtids \
 --start-datetime="2026-09-05 16:11:00" \
 --stop-datetime="2026-09-05 16:11:30" \
 /var/lib/mysql/mysql-bin.000037 > /tmp/db_recovery.sql
```

Restore the table data .

```
root@db-03:/tmp# mysql -u root -p -h localhost recovery < table_recovery.sql
Enter password:
root@db-03:/tmp#
```

After recover point in time .

```
mysql> select * from recovery_bank;
+----+---------------------+---------------------+
| id | name                | insert_date         |
+----+---------------------+---------------------+
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|  2 | ICICI               | 2026-09-05 16:07:54 |
|  3 | SBI                 | 2026-09-05 16:08:01 |
|  4 | BANK OF BARODA      | 2026-09-05 16:08:13 |
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+----+---------------------+---------------------+
5 rows in set (0.00 sec)

mysql> select * from recovery_bank;
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|  8 | Bank of India       | 2026-09-05 16:11:26 |
+----+---------------------+---------------------+
8 rows in set (0.00 sec)
```

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